

Three questions, one working pipeline

The ask

- 1 What is the market writing about?
- 2 Where is our own library thin?
- 3 Can we search our own work, with sources?

The data

Market (external): AG News

63,396 news items from 2,000+ sources

Our library (internal): CNN/DailyMail

306,913 full-length articles

Both sides filtered to tech news with the same rule, so the comparison is fair.

The outcome

370K documents processed

32 tech topics found and ranked by gap

7/7 right article found by the assistant

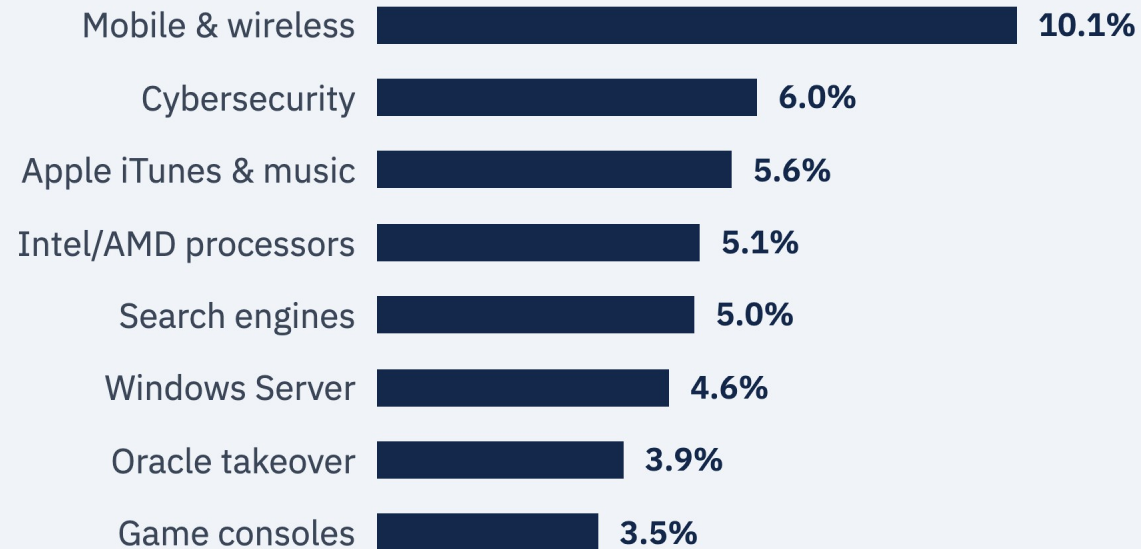
0 made-up answers in testing

HOW IT WORKS



Market tech news falls into 32 clear topics

Largest topics, as a share of market tech news



A leftover stock-market topic (not tech) is flagged and left out.

How I found them

31,007 tech news items, grouped by meaning with BERTopic

GPT-4o-mini gives each group a readable name

What I checked

Topics that were not tech

7 of 39 → 2 of 32

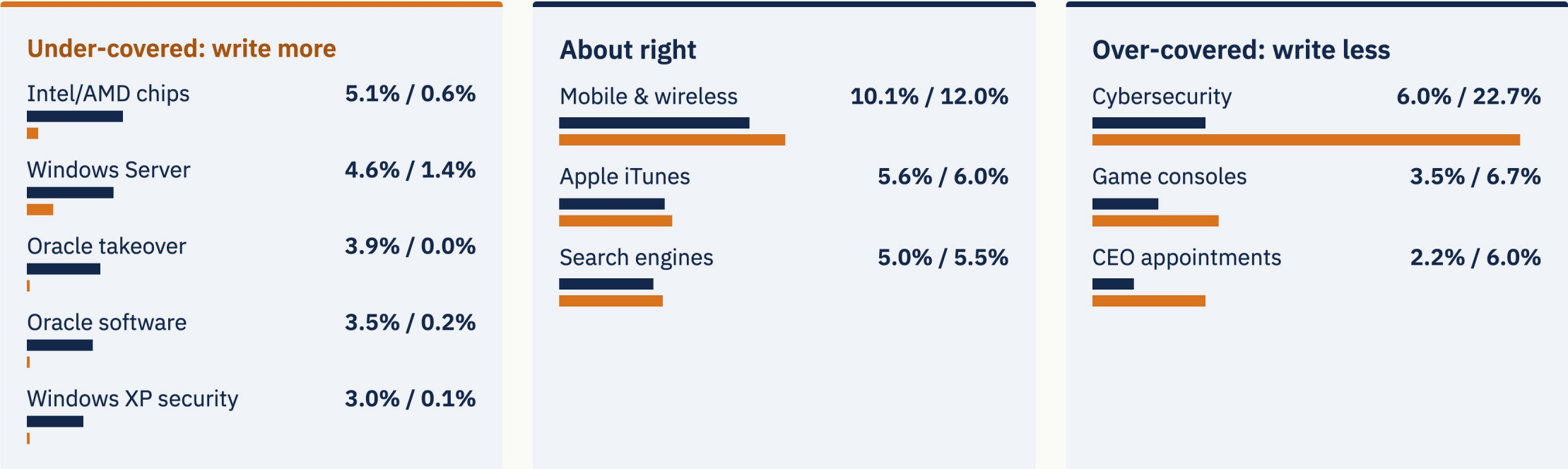
Rows with leftover HTML

6% → 0%

Biggest gaps: enterprise software and hardware

Same 32 topics on both sides · shares, not raw counts · match threshold 0.45, chosen after testing 3 values

Market (external share) Our library (internal share)



Some gaps are partly timing: the market news is older than our library. Treat this as a shortlist to review.

Answers from our own library, with sources



10 test questions · every answer checked by hand

7/7

found the right article in
the top 5

86%

of citations correct (6 of 7)

0/10

answers with made-
up facts

2/2

out-of-scope
questions declined

What I checked: caught the gap tool disagreeing with Part 3 and fixed it · fixed a crash on unanswerable questions · re-checked every label after the final run

Three moves for the content team

01

Write next

- Enterprise software series: Oracle, Windows Server, Linux and open source
- Chips and hardware: Intel/AMD, servers, storage
- Move some cybersecurity effort to these gaps

02

Invest in

- A current, dated news feed (e.g. GDELT) to spot emerging topics
- Connecting the firm's real report library
- A stronger tech filter and embedding model

03

Improve the assistant

- Keyword + vector search, plus a reranker
- Answer “how much do we have on X?”
- Re-ask automatically when a citation is missing
- A bigger test set, built with the content team

The top gaps share one theme: enterprise technology for business leaders.

Runs end to end on a laptop; saved results make reruns fast.